

VARANUS ISLAND, Australia

WMO: 953030

Lat: 20.656S

Lon: 115.576E

Elev: 80

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	65.9	66.8	43.2	41.2	70.8	45.3	44.8	71.3	38.3	70.2	35.5	68.6	20.4	170	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	7.5	92.2	77.5	90.6	78.0	89.4	78.1	83.9	85.9	82.4	85.7	81.3	85.5	12.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.5	175.6	84.9	81.7	165.4	84.1	80.3	157.6	84.1	48.3	85.7	46.5	85.5	45.3	85.4	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.7	32.0	28.7	64.7	97.1	0.5	2.2	64.3	98.7	64.0	100.0	63.7	101.2	63.3	102.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.7	83.3	84.2	85.3	83.3	78.3	73.4	71.3	72.4	74.2	77.7	79.8	81.6		
	DBStd	5.40	2.79	2.64	2.29	2.41	3.07	2.56	1.90	1.87	2.29	2.97	2.83	2.72		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	10476	1032	959	1094	999	876	701	660	694	727	860	893	981		
	CDD65	5001	567	539	629	549	411	251	195	229	277	395	443	516		
	CDH74	43662	6439	6494	7983	6324	3146	625	136	341	832	2536	3623	5183		
	CDH80	14032	2323	2622	3572	2216	450	4	0	3	52	425	828	1537		
(14) Wind	WSAvg	15.9	17.8	16.3	14.5	13.2	14.5	16.3	15.1	14.1	16.2	17.8	17.7	17.1		
(15) Precipitation	PrecAvg	10.7	0.9	1.9	2.0	1.3	1.5	1.6	0.7	0.2	0.0	0.0	0.1	0.3		
	PrecMax	27.0	3.9	8.3	10.4	11.1	7.3	8.2	5.2	0.8	0.3	1.0	1.0	1.7		
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	6.2	1.2	2.6	2.8	2.5	2.0	1.9	1.1	0.2	0.1	0.2	0.2	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.8	94.4	94.3	91.3	87.1	80.7	78.6	80.0	83.7	88.0	92.0	91.7		
		MCWB	77.0	79.1	77.3	76.4	72.3	68.8	68.2	68.6	68.9	72.9	72.1	75.9		
	2%	DB	90.9	91.7	91.7	89.6	85.3	79.2	76.5	78.3	81.5	85.5	87.8	89.4		
		MCWB	79.4	79.6	77.9	75.4	71.6	68.6	66.5	67.7	68.9	73.0	74.3	76.8		
	5%	DB	89.0	90.0	90.2	88.2	83.8	78.1	75.4	77.2	79.8	84.0	85.9	87.7		
		MCWB	79.4	79.8	78.4	75.3	71.3	67.8	65.8	66.7	68.3	72.5	74.7	76.6		
	10%	DB	87.5	88.6	89.0	87.1	82.5	77.1	74.4	76.0	78.2	82.4	84.3	86.2		
		MCWB	79.2	79.6	78.5	75.3	70.7	66.8	65.1	65.8	67.5	71.6	74.2	76.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	84.6	85.8	85.4	82.8	78.0	76.0	72.0	74.6	77.3	82.4	83.9	81.3		
		MCDB	85.9	87.5	86.9	86.1	80.4	77.9	75.0	76.2	78.5	83.8	85.0	86.3		
	2%	WB	82.8	83.8	83.3	81.1	76.5	73.6	70.6	72.4	73.1	77.5	79.4	80.0		
		MCDB	85.9	86.2	85.7	85.2	80.3	76.3	74.0	75.1	75.9	79.3	82.2	85.2		
	5%	WB	81.7	81.9	81.9	79.8	75.4	72.2	69.3	70.8	71.3	74.6	77.1	79.1		
		MCDB	85.8	85.8	85.6	84.8	80.6	75.5	73.4	74.0	76.0	80.6	82.0	84.4		
	10%	WB	80.6	80.8	80.9	78.5	74.3	70.7	68.0	69.1	69.9	73.2	75.7	78.1		
		MCDB	85.5	85.8	85.5	84.4	80.7	75.0	72.7	73.4	75.2	79.9	81.6	83.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	7.9	7.5	6.9	6.2	5.2	5.9	7.4	8.5	9.0	8.9	8.5		
		MCDBR	9.1	9.2	8.6	8.1	7.5	6.1	7.0	8.6	9.9	10.0	10.3	9.8		
	5% WB	MCWBR	4.0	4.1	4.6	5.3	5.6	5.3	5.1	5.0	5.4	5.0	5.2	4.4		
		MCWBR	7.9	8.3	8.1	7.5	7.1	5.4	6.4	8.3	8.9	9.2	9.1	8.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.400	0.397	0.374	0.353	0.335	0.325	0.316	0.315	0.333	0.353	0.357	0.373		
		taud	2.539	2.563	2.623	2.644	2.638	2.652	2.657	2.619	2.564	2.529	2.549	2.572		
	Ebn at Noon	Ebn at Noon	300	297	296	291	284	281	288	298	304	307	311	308		
		Edh at Noon	35	34	31	28	26	25	25	28	32	35	35	34		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2429	2332	2075	1772	1475	1326	1446	1798	2136	2414	2582	2594		
	RadStd	RadStd	135	154	140	116	120	99	101	53	82	45	54	88		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page